

1. Unification

Unification is the process of **making two logical expressions identical by finding appropriate substitutions for variables**.

It is used in **first-order logic and automated reasoning** to match two predicates.

Definition (Exam Oriented)

Unification is a process in which variables in logical expressions are replaced with suitable values so that the expressions become identical.

Example

Consider two predicates:

$P(x, y)$
 $P(A, B)$

To make them identical:

$x = A$
 $y = B$

Substitution set:
 $\{ x/A, y/B \}$

After substitution both become:

$P(A, B)$

2. Resolution

Resolution is a **rule of inference used in propositional and first-order logic** to prove whether a statement is true or false.

It works by **combining two clauses to eliminate a variable and derive a new clause**.

Definition (Exam Oriented)

Resolution is an inference rule used in artificial intelligence to **derive a new clause by eliminating complementary literals from two clauses**.

Example

Clause 1:

$P \vee Q$

Clause 2:

$\neg P$

By resolution:

Q

Here **P** and **$\neg P$** cancel each other, leaving **Q**.

Short Difference

Term	Meaning
Unification	Process of making two logical expressions identical by substitution
Resolution	Inference rule used to derive new conclusions from logical clauses